
EDUCATION

University of Wisconsin Madison BS Computer Engineering + Mathematics Certificate Expected 2030
Relevant coursework: CS 200, ECE 252, ECE 210, Math 234 2025**Harvard DCE** Math E16, Grade A**Aitchison College**

- Honourable Mention, 66th International Math Olympiad (IMO), Australia 2025
- School Rank 1/300. Led student council as Head Boy for a student body of over 3,000 students
- Best candidate Nationwide in Computer Science, and Environmental Management

Distinctions: Head Boy, Recipient Pakistan Blazer for Robotics, Founder of Innovation Team, Captain Basketball Team, Best Tent Pegger 2024

TECHNICAL SKILLS**Embedded & Hardware:** PCB design, embedded systems, firmware, Arduino, sensor integration (IMU, accelerometer, temperature, ultrasonic, IR) 2023**Programming:** Python, C++, Java, HTML**AI / ML:** RAG, LangGraph, vector databases, RLHF/DPO, model training

WORK EXPERIENCE**Children's Hospital of Philadelphia, USA- Technical Advisor (Wearable Prototyping)** Aug 2024 – August 2026

- Designed and built the MVP prototype for a pediatric remote patient monitoring wearable, integrating IMU, accelerometer, and temperature sensors to track sleep and fitness with >90% accuracy in step and sleep-cycle detection
- Developed and fabricated a custom PCB that cut component costs by 60% and enabled 24+ hours of continuous monitoring, creating a scalable foundation for clinical testing

Naseeb Online, USA – Intern

Aug 2025– Dec 2025

- Worked on LLM-enabled conversational systems used inside production recruitment workflows
- Built expertise in RAG architecture, vector databases, model training (RLHF and DPO), LangGraph
- Implemented conversational AI systems, integrating long term and short term memory with adaptive behavior

Medvive Enterprises B.V., USA – Intern

June 2024 – August 2024

- Enhanced minimal viable product of a web application through Java programming, debugging, and testing with Technical University of Delft

Punjab Information Technology Board, Lahore, Pakistan- Software Intern

June 2023 – August 2023

- Joined as a product development intern on PAITS project and enhanced skills in coding and project management

ENTREPRENEURIAL INITIATIVES**The Tech School- Founder, Curriculum Designer and Instructor** 2021-Present

- Founded and taught an inclusive Python program for differently abled vocational learners through LABARD, in partnership with Harvard Spark
- Built curriculum in programming, data science, and digital literacy to help students develop practical tech and freelancing skills

Epilet Watch- Designer & Developer

2023-2025

- Designed a wearable seizure-detection device using sensors and algorithms, with input from medical professionals and neurology experts
- Focused on creating a practical, accessible alert system to improve safety for people living with epilepsy

Pak Maweshi- Founder

2022-Present

- Built a livestock-focused agritech platform based on farmer interviews to address productivity and animal-management challenges.
- Secured PKR 900,000 through Spark Tank and NSIT awards; later supported Maweshi Madadgar, a women-centered livestock program backed by PKR 9 million in funding.

Diabetic Foot Analyzer- Creator

2021-2023

Created an Arduino-based device using sensors to monitor diabetic foot conditions and help detect early signs of complications.

RESEARCH PUBLICATIONS & NEWSPAPER ARTICLES

- “Decolonizing the Dataset through Contrastive Learning: An Emerging Perspective on Climate Resilience”, Submitted (2026)
- “Navigating Uncharted Waters: Artificial Intelligence, Machine Learning, and Labor Market Dynamics”, AI Journal (2025)
- “Innovation in Epilepsy Management in South Asia: Design and Development of a Wearable Seizure Detection Device”, J. of Dawn. R. (2024).
- “Privacy Considerations in Medical Technology: Role of Federated Learning and Differential Privacy in Wearable Devices”, JHSR Vol. 6 Issue 4 p. 17-21 (2024)

CONFERENCE PRESENTATIONS & Articles

- “Therapeutic Social Robot for Pediatric Male with Autism”, *12th International Young Researchers' Conference (IYRC WINTER 2025)*
- “From Sparks To Startups: Young Innovators And The Future Of Incubation”; *Guardians of the Future: Shaping Tomorrow with Generative AI, School of Tomorrow, 2024*
- “Privacy Considerations in Medical Technology: Role of Federated Learning and Differential Privacy in Wearable Devices” *International High School Student's Research Symposium, American Chemistry Society, 2023*
- Newspaper op-eds: Daily Times (2023), Pakistan Today (2022, 2021) on *environmental sustainability* and *robotics education*

ROBOTICS COMPETITION**World Robotics Olympiad**

- Team Lead for Pakistan at three WRO International Finals (Panama 2024, Germany 2022, Hungary 2019); 2019-2024
- Winner of autonomous car competition at the regional and national level
- *Built a self-driving car to complete the track in three laps while identifying and catering to traffic signals completely autonomously in the lowest time.*